Southwest CCDC 2024 Regionals

Blue Team Packet

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2 About the Competition

2.1 Game Time

The 2024 SWCCDC Regional game will be held on March 22nd, 23rd, and 24th 2024 at the University of Tulsa.

2.2 Team Organization

2.2.1 Gold Team

Competition Director: Gavin Bauer

The Gold Team directs and judges the competition. A decision from the Gold Team is final. Gold team is not part of the game scenario. Gold team members are to be inherently trusted.

2.2.2 Black Team

Black Team Co-Leads: Brady Deetz, George Louthan, and Peter Gehres

As the network and game operations team, members of the Black Team ensure that the network and systems are functioning as intended. Black team members are to be inherently trusted.

2.2.3 White Team

White Team Lead: Taylor Smith

White Team operates the business aspects of the competition. They provide competition support and manage the game scenario, grade injects, and perform other important competition tasks. White team members are to be inherently trusted.

2.2.4 Red Team

Red Team Lead: Jacen Kohler

The Red Team is a live adversary responsible for attacking blue teams and attempting to infiltrate systems and take services offline. They will start attacking only when the game starts, and have no knowledge of the specific hosts, OS, injects, or other game information.

2.2.5 Purple Team (Nationals calls this Orange team)

Purple Team Lead: Patrick Albert (CEO)

The Purple Team may play specific employees, affiliates, or other characters in the scenario. They may interact with IT systems as users, or with teams, and their actions may score services or score injects. Purple team members are allowed to be wrong, or lie, depending on how their character would behave in the competition scenario.

2.2.6 Blue Team

That's you! The blue team is tasked with operating and securing the network and performing business operations within the scenario.

2.2.7 Green Team

The Green Team is composed of sponsors who have observation rights for the game. A majority of the members of the Green Team are representatives of companies with open positions and we encourage you to chat with them. Anyone with a Green badge has the right to enter your team's room, however they should not interact with you or be distracting. You have the right to ask them to leave your room if you feel they are distracting but we encourage you to accommodate them if you're able to.

2.3 Trusted Teams

SWCCDC strictly respects the concept of **trusted teams**. In this region, the **Gold team**, **Black team**, and **White team** are trusted. Members of these teams can always be identified either by their physical badges (regionals) or their roles on discord (qualifiers).

Trusted team members will never lie to, or intentionally mislead, competitors. There may be questions they are not allowed to answer, or issues they aren't sure about, and they may occasionally make mistakes; but they will never lie to blue teams.

Trusted team members are completely out of scope for the red team to impersonate, explicitly or implicitly. Any accounts, files, or systems that appear to reference gold, white, or black teams are part of the competition infrastructure (and out of scope for red team).

Green Team members are also out of scope for the Red Team. While not a "trusted team" as they are not competition staff, Red Team is prohibited from impersonating them.

2.4 Scoring

2.4.1 Injects

Approximately half of the points come from the successful and on-time completion of the business and technical injects. Take great care in understanding who the target audience is and what is being asked. It is ok to ask for clarifications for questions you have about an inject. Answering the right question is crucial for gaining points. Any inject submitted after the deadline,

but on the same day, will receive a 50% late penalty. Injects from previous days cannot be submitted.

There are generally two types of injects:

Business Tasks - Things like:

- Write an incident response plan or policy
- Prepare an incident report for something that the red team did
- Assess the impact of some new cyber security or privacy regulation

Technical Tasks - Things like:

- Add TLS to a web server
- Implement firewall rules
- Implement an AD GPO to change the desktop background on all workstations

2.4.2 Services

The other half of the points are earned by keeping the scored services up throughout the game. The services are scored by the scoring engine which runs checks frequently. Every successful check will gain your team points, and every failed check does not. Service status is displayed on Portal's home page.

For the regional game we use Service Level Agreements (SLAs) and apply penalties if they are not met. If a single service is down for 4 consecutive checks, the SLA penalty would be applied and you will lose points. The amount of points lost is generally equal to the amount of points the service is worth. If you have 3 consecutive failed checks and the 4th check passes, the SLA will not apply. If you have 6 consecutive failed checks only 1 SLA penalty is applied as after 4 checks the counter is reset to 0. 8 consecutive failed checks would result in 2 SLA penalties.

2.4.3 Red Team

Every successful attack by the red team against a blue team causes the blue team to lose points. These attacks include gaining access to machines, creating persistence, and stealing confidential data.

Blue teams can regain up to half of the points lost if they detect the intrusion and file an incident report. Incident reports need to include specific information including:

- How the red team initially got access
- What level of access did they have (user account, system access, etc)
- A timeline of events
- Impact of the incident (website went down, credentials were compromised, etc)
- How did you remove their access (if applicable, this is not required for a report)

Detailed reports that match the red team's description of events will earn up to half of the points lost back. If an event does not have any impact on your company or services, such as port scanning, do not report it as an incident. These reports are filed via Portal.

Red team is explicitly prohibited from impersonating Gold, Black, and White team members.

2.5 Team Environment

2.5.1 Team Rooms

Every team will be given their own room to compete in. Rooms are assigned at random during the opening ceremony. The laptops CCDC uses will be tagged with a CCDC tag, do not touch any computer or equipment not associated with the competition. Also do not tape any materials to the wall except with the tape we provide you. If you have any questions about equipment or modifying the room please ask any competition staff member.

In general teams are responsible for their rooms when the competition is active. You have discretion to apply reasonable rules for access control and have the power to ask any non-competition staff member to leave. You are not permitted to physically touch any individual or intimidate any person in any way. If someone is not following the rules or is being a distraction and is not leaving, please reach out to the Gold Team. The Gold Team will determine what constitutes "reasonable rules."

In your room will be a judge, most often a coach from another team. They are there to ensure you are following the rules and should not interact with you.

You are permitted to play music at a reasonable volume in your room. The speaker and device playing the music should be off to the side and not connected to or nearby competition computers.

2.5.2 Team Coaches

Coaches are not allowed in the room their teams are competing in, and are not permitted to communicate with their teams during the competition in any way. They may monitor discord and Portal from another location.

2.6 Post Game

2.6.1 Nationals

The top team will advance to the National CCDC event in April. The second place team from all regions will compete in the wildcard competition, and the winner from that will also advance to nationals.

2.6.2 Feedback

We will provide feedback for how you did during the event on the 3rd day. This includes points received for each inject, service totals, and red team activity. Each of the teams will come to your room and debrief you on certain elements of the game. You'll have a chance to ask questions and get direct feedback more so than quals.

3 Scenario Description

3.1 Background

In qualifiers, blue teams worked for Galactic Garbage. That company has been acquired by the Galactic Empire, and the blue teams will work as the IT department at the office relocated to an Imperial space station under the organization name **Imperial Sanitation (IS)**, as the provider of garbage smashing and management services for that space station.

Galactic Garbage was a mid-sized company, with approximately 300 employees. They have a very small corporate HQ and cloud environment. Some of this has transferred over to Imperial Sanitation. In the business scenario, there are many other sites, field personnel, and a factory, but these may not be included in the competition IT simulation.

3.2 Additional Features

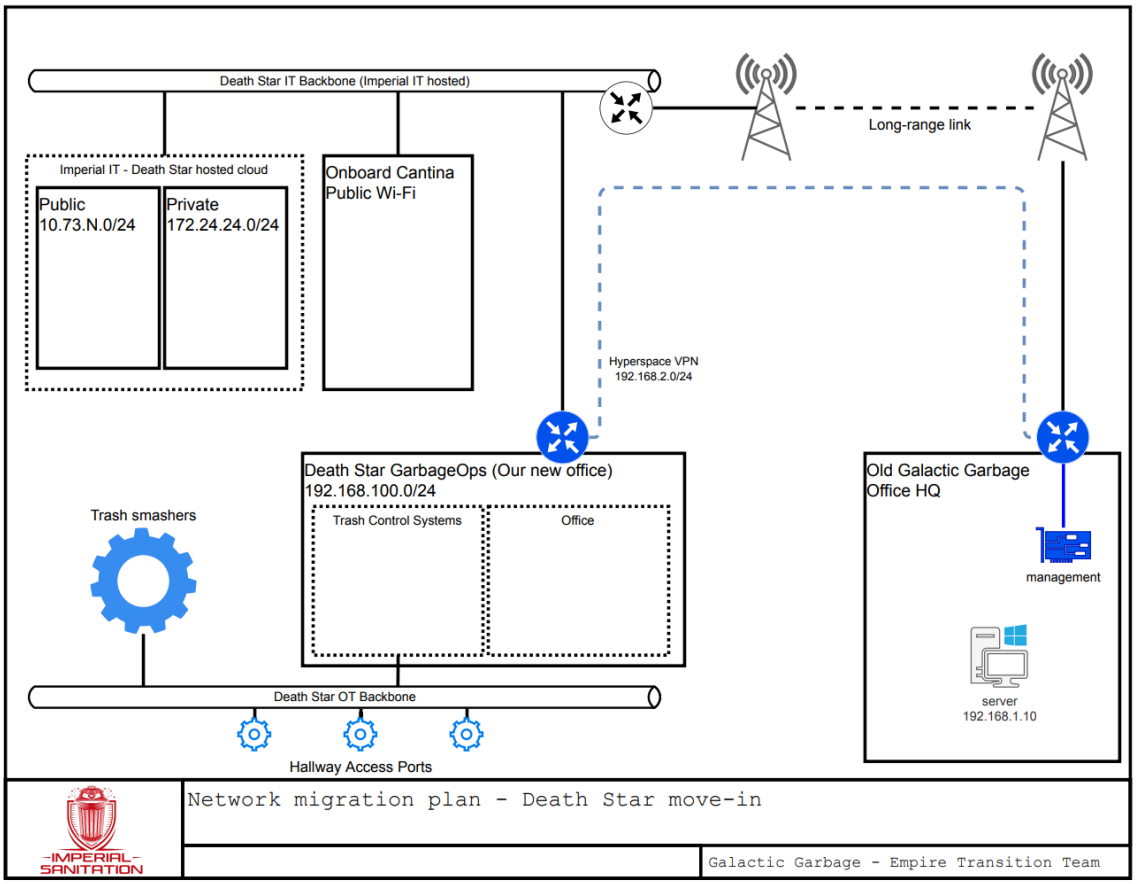
A social media site for the competition exists at https://blipper.swccdc.com/ for the purposes of threat and scenario intelligence. It's a lightly customized private Mastodon instance named after a social media company used as the scenario for a past competition. Red and Blue teams are allowed to create accounts, as are some observers (within limits) and competition staff, who may also post things in-character from people in the scenario.

Attacks against blipper are not allowed; however, it's a valid platform for social engineering.

3.3 Networking Specifics

The domain for this year's game is imperial.swccdc.com and your team domain is teamNN.imperial.swccdc.com (where NN is your 0-padded, two-digit, team number, like 01, 02, 03, … 09, 10, 11, etc.). A number of services fall under that domain. A network diagram is attached below with other important names and numbers listed on it.

3.4 Diagram



4 Competition Environment

4.1 SWCCDC Hosts

4.1.1 Computers in team rooms

During the regional event, you will only use SWCCDC provided computers. You should not bring any other computers into the competition rooms. Please take care when using our equipment.

4.1.2 Installing Apps

You are permitted to install any apps relevant to the competition.

4.1.3 Host Change Requests

All hosts must remain running the same major version of the operating system and software unless a host-change request has been approved. For example, a host running Ubuntu 22.04 and MySQL 5.7 must remain on 22.4 and MySQL 5.X unless a change has been approved. Changes are submitted via portal and must have valid business justification.

4.1.4 Limitations

Do not set any BIOS passwords, or other hardware/system level password outside of the operating system. We reuse these laptops every year, and need to have access to them. Setting the root/system password of the operating system is fine.

4.2 Networking & Switches

In your team rooms you will find many switches and networking equipment. If it’s marked as blue team equipment, it’s in scope for you to manage.

4.2.1 Black Team Equipment

Some items are Black Team equipment and out of scope for the red team. You are not permitted to login or modify any aspect of any device labeled “Black Team”.

4.3 OpenNebula (cloud)

4.3.1 Overview

OpenNebula remains an important item.

4.3.2 Credentials

Credentials will be delivered via Portal in the credentials section. You will not see them until the start of the game.

4.3.3 Snapshots

Details about snapshots and their limitations will be provided in the docs site (see section 4.8).

4.4 Portal

Portal is the central system you will use to see system status, receive and respond to injects, submit incident reports, and much more.

4.4.1 Accessing

Portal is at https://portal.southwestccdc.com and can be accessed from any computer. You do not need to be on the VPN or in the game station VM host to access Portal.

4.4.2 Registration

Team captains should register first, and then invite the rest of their teammates.

4.4.3 Inject Submissions

Injects must be submitted via the portal, no inject submissions sent via Discord, email, or other out of band system will be accepted. Put your response entirely in Portal, do not upload a document with a link to etherpad or any other document store. We only grade what we see in the inject response and will not follow any links to outside resources.

Only the most recent inject submissions will be graded. If you need to make a correction on something previously submitted, resubmit your entire response.

Portal allows you to mark a response as “Temp”, this means the inject will continue to show up as a “todo” item. Graders do not see whether a response is temporary or not, this is a way for you to manage your action items.

4.4.4 Coach

Your team coach is required to have an account on Portal associated with your school. They can use Portal to observe your team during the event.

4.5 Slack

Every room will have one computer with Slack installed on it, this is your primary way to communicate with competition staff.

4.5.1 Getting Help

Every team has a dedicated channel with their team number. Do not DM any competition staff; ask any questions in the team channels. The team channels only have competition staff and your user in them and are out of scope for the Red Team.

4.6 Document Collaboration

For inject response writing, you are welcome to use local tools like Word, Excel, their OpenOffice equivalents, or similar. If you wish to collaborate on writing, during the opening ceremony we will go over if Etherpad will be in scope, or give you an alternate approved collaboration mechanism.

4.7 Root Certificate

Many in-competition locations use TLS certificates signed by our internal certificate authority. In order to trust those endpoints from the machine you're using to connect to the VPN, the download package includes that certificate authority's root certificate. We highly recommend that, if you install that certificate to your root trust store for the competition, you remove it after the end of the game.

4.8 Docs Site

An internal documentation site will be available during the competition at https://docs.imperial.swccdc.com. This will host more current in-game information during the competition and may be referenced in injects and announcements. The link to the docs site will be provided during the competition as well. It is only available from the competition network.

4.9 Privacy

4.9.1 Monitoring

All activity on the competition network is subject to monitoring and recording. We will take measures to observe any network and computer activity to enforce the competition rules, general fairness, and acceptable use of the competition networks.

This may include logging any DNS queries and other traffic from any computer that you connect to the competition VPN.

4.9.2 Publishing

We may publicly publish any recorded information about teams internet usage during the competition. This includes but is not limited to urls, dns, searches, and time spent.

4.9.3 Credentials

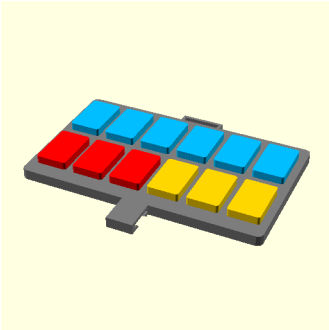
We strongly encourage the red team to not publish any user-specific information they may have gained during the competition; however we also advise blue teams to not use any “real” passwords for in-scope accounts. We are not responsible for the release of any credentials used during the competition.

4.10 ID Badges

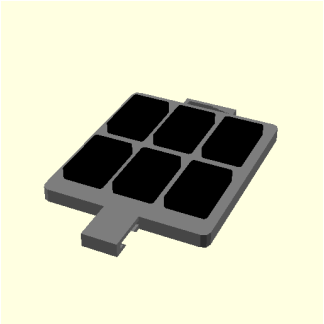
For 2024 Regionals, team members will be issued an ID badge and holder corresponding to their team.

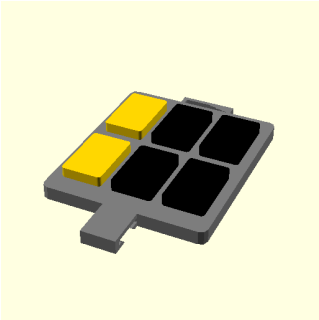
4.10.1 Competition Role - Badge Holder

All participants in Southwest CCDC will be issued a badge holder. These designate the role they play during the competition. Badge holders representing Trusted teams are out of scope for impersonation or counterfeiting.

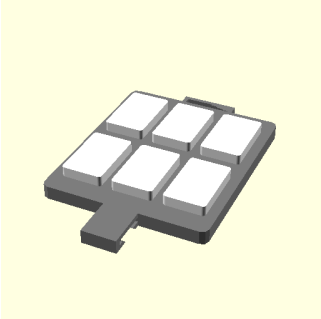
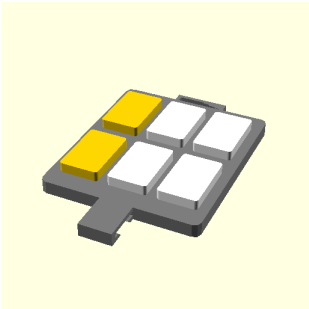


Gold team (competition director) - may not be impersonated

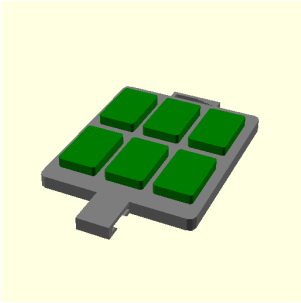




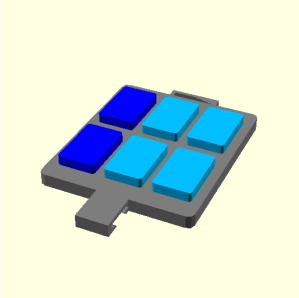
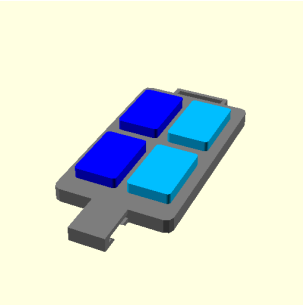
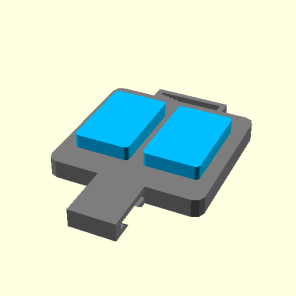
Black team and Black team leads - may not be impersonated

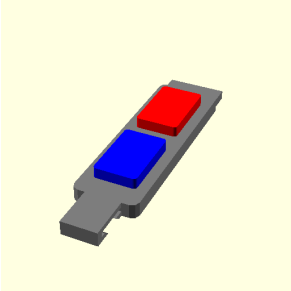
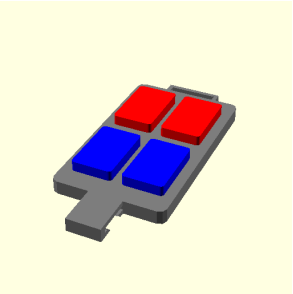
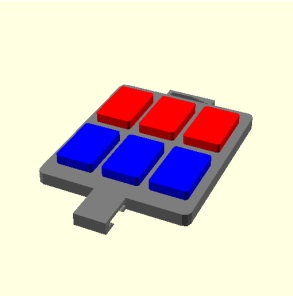
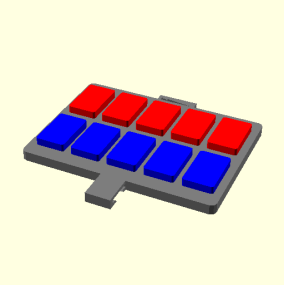
White team and white team leads - may not be impersonated



Visitor/observer/sponsor - may not be impersonated

Blue team coach, captain, and team member - may not be impersonated

In-character badge holders - impersonation may be allowed

4.10.2 Scenario Role - ID Card

In addition to the badge holder (which represents a person’s role in the competition), some or all participants may be issued ID badges to hang from them. These are entirely

in-character/in-scenario, and red team has no restrictions around forging them. Where the badge holder shows someone's role in the competition, the ID card shows their character's role in the simulated company.

5 Event Information

5.1 Schedule

Thursday, March 21st

- 3:00 pm - Check-in opens at hotels

Friday, March 22nd

- 7:00 am Registration opens
- 8:00 am Day 1 briefing
- 9:00 am Game start
- 5:00 pm Game stop
- 5:15 pm Day 1 debrief
- 6:00 pm Fun event at Circle Cinema

Saturday, March 23rd

- 8:30 am Day 2 briefing
- 9:00 am Game start
- 5:00 pm Game stop
- 5:15 pm Day 2 debrief
- 5:30 pm Job fair and dinner, Allen Chapman Student Union

Sunday, March 24th

- 8:00 am Building opened
- 8:30 am Day 3 briefing
- 9:00 am Individual debrief & room teardown
- 1:00 pm Presentations
- 2:00 pm Awards and closing ceremony
- 3:00 pm End of competition

Monday, March 25th

- 11:00 am Check-out of hotels

5.2 Locations

The SWCCDC event is held at Helmerich Hall at the University of Tulsa.
The full campus map can be found at:
https://utulsa.edu/wp-content/uploads/2024/01/24016_ADA_CampusMap_4jan.pdf

5.2.1 Event Location

Please park in the UMC Lot, there will be CCDC signage to confirm you are in the right lot. The competition is in the building just across the street from the parking lot.

5.3 Hotel Information

We will be using two hotels for this year, the Fairfield Inn & Suites Tulsa Downtown and the Courtyard Tulsa Downtown. We will do our best to keep teams in the same building but we cannot guarantee this. Both hotels will provide breakfast for free.
Fairfield address is: 111 N Main St, Tulsa, OK 74103
Courtyard address is: 415 S Boston Ave, Tulsa, OK 74103

Teams will need to organize their own transportation between CCDC events.

6 Competition Rules

6.1 Competitor Eligibility

6.1.1 Full Time

All students must be considered full time by their institution. A student in their final semester before graduating is exempt from this requirement under certain circumstances. Please reach out to the competition director for details.

6.1.2 Aging Out

Students may only participate in up to 5 SWCCDC seasons. Participation in any event, even only the qualifying event, counts towards a CCDC season.

6.1.3 Team Limit

Students may only participate on one SWCCDC team per season, and may not participate in any role other than the blue team without explicit approval from the competition director.

6.2 Team Composition

6.2.1 Players

Teams are composed of up to 12 students on a roster. From this roster they may select up to 8 students to be “active players”. Only active players can participate in a CCDC event.

Student Category	Minimum Number	Maximum Number
Students on Roster	1	12
Active players	1	8
Graduate Students on Roster	0	2

6.2.2 Team Rosters

After the registration deadline no additions to the team roster are permitted.

6.2.3 Substitutions

Between events teams may substitute an active player with a backup player, however once an event has started no substitutions may be made until the event is over. For example, between

Quals and Regionals you are free to change who is going to play, however you cannot swap a player between day 1 and day 2 of regionals.

6.2.4 Coach

All teams must have a coach who is a staff or faculty member of the university the team is representing. In the event your coach is not associated with your university you must provide a university contact that is either a staff or faculty member.

6.2.5 Team Captain

Teams must select one person to be a team captain. Team captains are the liaison between competition staff and the team. Teams may select one or more co-captains as well.

6.2.6 Bringing Team Members

Please only bring the team members and coach who are competing in the competition. We are not able to provide hotel accommodations for any backup team members and backup team members are not allowed in the competition building.

6.2 Team behavior

6.2.1 Network Scanning

Blue teams are authorized, and encouraged, to scan networks they have full control over. These are outlined in section 3.

Scanning the simulated internet, or any other team's networks is strictly prohibited.

6.2.2 Attack Response

Teams are allowed to use active response mechanisms such as TCP resets when responding to suspicious/malicious activity. Any firewall rule, IDS, IPS, or other defensive action that prevents the scoring engine or manual scoring checks from succeeding are exclusively the responsibility of the teams.

6.2.3 Coach Limitations

During the qualifying event coaches are not allowed to:

- Enter team rooms while they are competing
- Communicate with the team in any way while they are competing including between days
- Interact with other teams except in their capacity as a room judge
- Participate on the Red, Gold, Black, White, or Purple/Orange team

6.2.4 Other Teams

Teams may not enter any other team's space, and may not interfere with another team competing. Teams are prohibited from accessing or attempting to access any system of another team. This includes using another team's credentials on a shared service. Teams are also not allowed to impersonate any other team or competition staff.

6.2.5 System Access

White/Black Team members will occasionally need access to a team's systems for scoring, troubleshooting, etc. Teams must immediately allow Black and White Team members' access when requested.

The Black Team also may place accounts named "blackteam" onto systems to allow for persistent access, including scoring. This may also include root SSH keys with "blackteam" in the comment field. Teams must not remove, change the password, change the permissions, or otherwise modify any Black Team account or credential. In the event of an accidental change, notify Black Team immediately and we will attempt to work with you to restore access.

Failure to follow this section will result in the inability to score services or injects. If we are unable to access a system for scoring we will score as if the service is down or the inject was not completed.

6.2.6 Competition Environment

Teams must not connect any devices or peripherals to the competition network unless specifically authorized to do so by Black or White Team members.

Teams may not remove any item from the competition area unless specifically authorized to do so by Black or White Team members including items brought into the team areas at the start of the competition.

6.2.7 Electronic Devices

Flash drives, removable media, tablets, and similar devices are not permitted in the competition room. Do not bring in electronic devices and peripherals including laptops, keyboards, mice, etc. Phones cannot be used while you are actively competing, however you are allowed to step out to take a phone call or send a text if you need to. You can also use phones to play music.

You are permitted to play music at a reasonable volume from a speaker set away from the game systems if desired.

6.2.7 Permitted Outside Materials

Printed reference materials (books, magazines, checklists) are permitted in competition areas and teams may bring printed reference materials to the competition.

6.2.8 Hardware

Teams may not modify the hardware configurations of competition systems. Teams must not open the case of any server, printer, PC, monitor, KVM, router, switch, firewall, or any other piece of equipment used during the competition. All hardware related questions and issues should be referred to the Black Team.

6.2.9 Outside Help

Teams must compete without “outside assistance” from non-team members including their coach from the start of the competition to the end of the competition (including overnight hours for multi-day events). All private communications (calls, emails, chat, texting, directed emails, forum postings, conversations, requests for assistance, etc) with non-team members including a coach that would help the team gain an unfair advantage are not allowed and are grounds for disqualification and/or a penalty assigned to the appropriate team.

6.2.10 Red Team

Team members will not initiate any contact with members of the Red Team during the hours of live competition. Team members are free to talk to Red Team members during official debrief events.

6.3 Internet Usage

6.3.1 Online Resources

In general, any online resource such as FAQs, how-to, forms, public company websites can be used for competition events provided:

- There is no fee to sign up
- Access is not based on previous membership/fee
- Access is available to all CCDC competitors immediately
- A CCDC team member did not create the resource

Examples of authorized resources include:

- Stack overflow forms
- Cisco support pages

Examples of unauthorized resources include:

- Public github repository in the team captain’s user
- Blog post on a private university website
- A support form that requires a monthly subscription

The only two exceptions to this rule are your Portal account and a Discord account, both of which you need to make before the game starts.

If you have any doubts or questions on if a resource is permitted or not, contact the competition director.

No Peer to Peer file sharing is authorized unless explicitly allowed by the competition director.

6.3.2 Online Accounts

Students are prohibited from logging into any account created before the competition. E.g. You are not allowed to login to your personal Gmail, even if it's just to accept an invitation to something else.

For any system that requires an account, you will need to make a new account during the competition.

6.3.3 Online Collaboration

We will provide a mechanism to support online collaboration on word documents, we will go over what this is during opening the ceremony.

6.3.4 Scripts

Any pre-staged custom scripts are not allowed to be used during the regional event unless they are in the main branch of this repo: https://github.com/SouthwestCCDC/2024-Regionals-Shared. Even if in a public place such as a university hosted server or a public github repository, scripts are not allowed in digital form. Any request to add scripts to the repo must be made in the specified timeline. All scripts in the repo can be used by any team.

You CAN print out any script/config/LaTeX template and bring it into the room on paper and type it by hand into a host on the environment. Scanners/OCR/other digitizing is not allowed for any printed out materials.

6.3.5 LLMs and Generative AIs

LLMs such as Chat GPT are allowed during the environment provided the models are:

- Freely available to any person (e.g. no Chat GPT 4)
- Not invite-only
- Commonly found
- Not trained for CCDC or competitions in any way

In general Chat GPT 3.X is fine, for any other models we encourage you to reach out to see if they are authorized.

6.4 Professional Conduct

6.4.1 Personal Conduct

All participants, including competitors, coaches, White Team, Red Team, Ops Team, and Gold Team members, are expected to behave professionally at all times during all CCDC events including preparation meetings, receptions, mixers, banquets, competitions and so on.

Activities such as swearing, consumption of alcohol or illegal drugs, disrespectful or unruly behavior, sexual harassment, improper physical contact, becoming argumentative, willful violence, or willful physical damage, and general harassment have no place at the competition and will not be tolerated.

6.4.2 Host Site Rules

In addition to published CCDC rules, Host Site policies and rules apply throughout the competition and must be respected by all CCDC participants.

6.4.3 Alcohol

All CCDC events are alcohol free events. No drinking is permitted at any time during competition hours.

6.5 Questions, Disputes, and Disclosures

6.5.1 Questions

Prior to the competition team captains are encouraged to work with the Competition Director and their staff to resolve any questions regarding the rules of the competition or scoring methods before the competition begins.

6.5.2 Disputes

During the competition protests by any team must be presented in writing by the Team Captain to the Gold Team as soon as possible. The competition officials will be the final arbitrators for any protests or questions arising before, during, or after the competition. Rulings by the competition officials are final. All competition results are official and final as of the Closing Ceremony.

6.5.3 Disqualification

In the event of an individual disqualification, that team member must leave the competition area immediately upon notification of disqualification and must not re-enter the competition area at any time. Disqualified individuals are also ineligible for individual or team awards.

In the event of a team disqualification, the entire team must leave the competition area immediately upon notice of disqualification and is ineligible for any individual or team award.

6.6 Violations

6.6.1 Reporting

Any person observed violating these rules should be immediately reported to the competition director.

6.6.2 Enforcement

Competitors behaving in an unprofessional manner may receive a warning from the White Team, Gold Team, or Black Team for their first offense. For egregious actions or for subsequent violations following a warning, competitors may have a penalty assessed against their team, be disqualified, and/or expelled from the competition site. Competitors expelled for unprofessional conduct will be banned from future CCDC competitions for a period of no less than 12 months from the date of their expulsion.

7 More Information

If you have any questions about whether you can do something or not, please ask! info@southwestccdc.com is the fastest way to get a question answered.